# CAUSALITY ASSESSMENT OF DRUG REACTIONS

Task 3: Evaluating the relationship between a drug and an observed adverse reaction

| 1. CASE STUDY | |
|---|---|
| **Patient** | 45-year-old female |
| **Suspected drug** | Amoxicillin |
| **Adverse reaction** | Skin rash |
| **Onset** | 3 days after starting the drug |
| **Outcome** | Recovered after stopping the drug |

**2. ASSESSMENT CRITERIA**

| No. | Criterion | Question | Finding | Result |
|---|---|---|---|---|
| 01 | **Temporal relationship** | Did the reaction occur after the drug was administered? | The rash appeared 3 days after amoxicillin was started, a plausible time for a hypersensitivity reaction. | **Yes** |
| 02 | **Dechallenge** | Did the reaction improve after stopping the drug? | The patient recovered after the drug was withdrawn (positive dechallenge). | **Yes** |
| 03 | **Alternative causes** | Could the reaction be explained by another cause? | No other drug, disease or infection is reported that could explain the rash. | **No** |
| 04 | **Rechallenge (if applicable)** | Did the reaction reappear on re-administration? | Amoxicillin was not given again, and rechallenge is not ethically advisable after an allergic reaction. | **Not done** |

**3. CLASSIFY THE CAUSALITY (WHO-UMC)**

| Category | Meaning | This case |
|---|---|---|
| **Certain** | Clear evidence of a causal relationship, including a positive rechallenge. | No (rechallenge not done) |
| **Probable / Likely** | Likely relationship: plausible timing, improves on withdrawal, other causes unlikely. Rechallenge not required. | **YES** |
| **Possible** | Relationship cannot be ruled out, but other causes could also explain it. | No |

**4. SUPPORTING SCORE (NARANJO SCALE)**

| Question | Answer | Score |
|---|---|---|
| Previous conclusive reports of this reaction? | Yes (rash is a well-known reaction to amoxicillin) | +1 |
| Did the event appear after the drug was given? | Yes | +2 |
| Did it improve when the drug was stopped? | Yes | +1 |
| Did it reappear on re-administration? | Not done | 0 |
| Alternative causes that could explain it? | No (none reported) | +2 |
| Placebo, drug level, dose-response, previous similar reaction, objective confirmation? | Not available / not reported | 0 |
| **Total score** | **5 to 8 = Probable** | **6** |

**5. CONCLUSION**

The reaction is classified as **PROBABLE (Likely)**. The rash began 3 days after starting amoxicillin, resolved after the drug was stopped, and no alternative cause is evident. It is not classified as Certain because rechallenge was not performed. The patient should avoid amoxicillin and other penicillins in future, and the allergy should be recorded in her medical file.

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